Real-Time, Vision-Based Biomechanical Form Assessment for Resistance-Training Exercises

Problem Statement

1. Background

Recent advances in computer vision and human pose estimation have enabled automatic monitoring of resistance-training exercises using only a single RGB camera, eliminating the need for wearable sensors, depth cameras, or multi-camera systems. Existing approaches based on MediaPipe pose estimation, Temporal Convolutional Networks (TCNs), and RepNet have demonstrated reliable real-time exercise recognition and repetition counting under realistic gym conditions using ordinary hardware. However, these systems remain **form-agnostic**, treating every detected repetition equally regardless of whether it was executed with correct or incorrect biomechanics. Consequently, a reported repetition count reflects only exercise quantity, without providing any indication of movement quality or execution safety.

2. Motivation and Injury Statistics

2.1 Resistance Training Injury Rates and Mechanisms

Resistance training is widely recognized as an effective method for improving physical health and performance; however, incorrect exercise execution can result in significant musculoskeletal injuries. Epidemiological studies report that resistance training and bodybuilding activities are associated with an injury incidence rate ranging from **2.4 to 5.3 injuries per 1,000 hours of training**.

For an individual performing approximately **4 hours of resistance training per week (~200 hours annually)**, these injury rates indicate a considerable probability of experiencing a reportable training-related injury over time, particularly when improper movement mechanics, excessive loading, and insufficient technique awareness are present.



Figure 1: Reported resistance-training injury incidence rate range (2.4–5.3 injuries per 1,000 training hours)

At the gym-community level, these injury rates represent a substantial problem. For example, a medium-sized gym with approximately **50 active lifters** may experience dozens of training-related injuries annu-

ally, particularly during high-load compound movements such as **deadlifts, squats, and bench presses**, where biomechanical errors can significantly increase joint stress.

Previous studies have identified several major contributors to resistance-training injuries, including:

- **Technical execution errors**, such as incorrect body alignment and improper joint positioning.
- **Fatigue and excessive loading**, which often lead to deterioration of movement quality.
- **Insufficient exercise knowledge and lack of professional supervision.**

Among reported injury mechanisms, technical errors have been associated with approximately **31% of injuries**, while fatigue or overloading has been reported in up to **81% of cases**. Additionally, dropping weights represents a major mechanism of acute injuries, accounting for approximately **65.5% of reported incidents**.

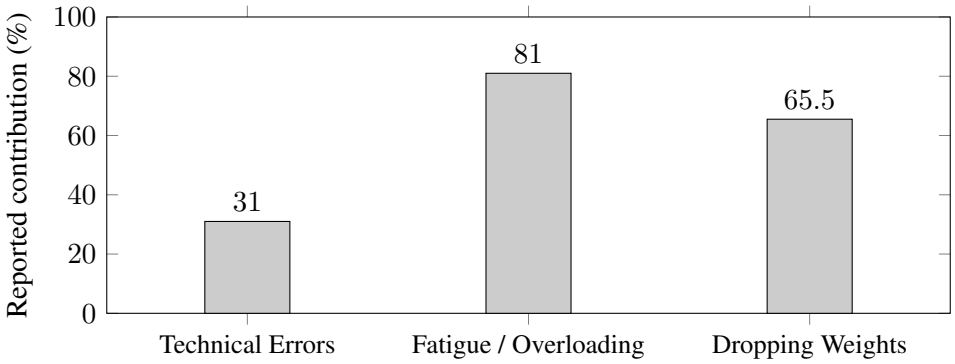


Figure 2: Reported contribution of major injury mechanisms in resistance training

The most frequently affected anatomical regions include:

- **Shoulder injuries:** caused by incorrect pressing mechanics, excessive loading, and poor shoulder positioning.
- **Lower back injuries:** commonly associated with spinal rounding during deadlifts and squats.
- **Knee injuries:** frequently resulting from biomechanical deviations such as knee valgus during squat movements.

These injuries highlight the importance of detecting incorrect movement patterns early and providing immediate corrective feedback during training.

2.2 Member Retention and Absence of Personal Guidance

Despite the growing popularity of fitness facilities, maintaining consistent exercise habits remains challenging for many gym members. Surveys indicate that approximately **60% of new gym members discontinue training within the first six months**.

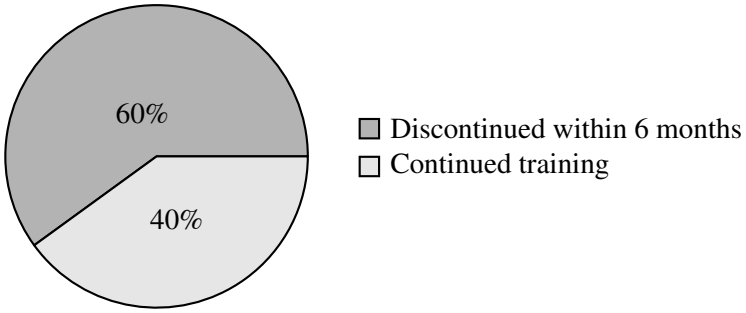


Figure 3: New gym member training status within the first six months

A major contributor to this dropout rate is the absence of continuous professional guidance. Many beginners experience:

- Fear of injury due to incorrect exercise execution.
- Lack of confidence when using unfamiliar equipment.
- Difficulty accessing personal trainers because of financial constraints.

Traditional personal coaching can provide effective feedback; however, it is often unavailable or expensive for many users. Therefore, an automated AI-based coaching system can provide scalable and affordable guidance.

Furthermore, many users show reluctance toward wearable sensing devices because of discomfort, maintenance requirements, and the need to wear specialized hardware during workouts. This reinforces the need for **non-intrusive, camera-based exercise assessment systems** that can provide feedback without affecting the user's training experience.

3. Research Gap

Several studies have attempted to evaluate exercise execution quality using computer vision, biomechanical analysis, or multimodal sensing. While these approaches demonstrate promising performance, each sacrifices one or more requirements necessary for practical deployment in everyday gym environments.

FLEX (Yin et al., 2025) is capable of detecting biomechanical errors such as knee valgus and spinal rounding while providing explainable corrective feedback through a Fitness Knowledge Graph. However, the system depends on **five synchronized RGB cameras together with surface electromyography (sEMG) sensors**, making it unsuitable for low-cost deployment in commercial gyms.

Fitness-AQA (Parmar et al., 2022) performs workout form assessment using real-world weightlifting videos and expert annotations. Nevertheless, it produces only a **binary correct/incorrect assessment** for an entire exercise clip, without identifying the specific biomechanical errors responsible for poor execution, and it is not designed for real-time feedback.

CoT-AFA (Qi et al., 2025) introduces explainable fitness assessment by generating chain-of-thought reasoning for detected form errors. Although it improves interpretability, the framework has not been evaluated for **real-time single-camera deployment**, limiting its applicability during live workouts.

M3GYM (Xu et al., 2025) provides one of the largest datasets recorded under unconstrained gym conditions and includes expert action assessments. However, the dataset is captured using **eight synchronized camera views**, making it impractical for real-world gym installations that typically rely on a single surveillance camera.

In the rehabilitation domain, datasets such as **KIMORE** and **UI-PRMD** demonstrate that movement

quality can be estimated from single-camera recordings. However, these datasets focus on slow rehabilitation exercises rather than resistance training and typically generate only a **continuous quality score** instead of identifying individual biomechanical errors or producing actionable corrective feedback.

Although these systems address individual aspects of exercise assessment, **none simultaneously combines**:

- Real-time exercise recognition.
- Real-time repetition counting.
- Single ordinary RGB camera.
- No wearable or EMG sensors.
- Detection of individual biomechanical form errors.
- Explainable corrective feedback.
- Valid vs. invalid repetition counting.
- Validation under realistic, unconstrained gym conditions.

Consequently, a fundamental gap remains between existing laboratory-oriented fitness assessment systems and the practical requirements of an intelligent gym assistant that can provide real-time, explainable coaching using only a single ordinary camera.

Table 1: Comparison of existing exercise-assessment systems against the proposed system

System	Single Camera	No Wearables	Real-time	Per-error Detection	Explainable Feedback	Valid/Invalid Rep. Count	Real Gym
Base Thesis (Cridlig)	✓	✓	✓	×	×	×	✓
Fitness-AQA	~	✓	×	×	×	×	~
FLEX	×	×	×	✓	✓	×	×
CoT-AFA	✓	✓	×	✓	✓	×	—
M3GYM	×	✓	×	~	~	×	✓
KIMORE / UI-PRMD	✓	~	×	×	×	×	×
Proposed System	✓	✓	✓	✓	✓	✓	✓

✓ = Fully satisfies the criterion. ~ = Partially satisfies the criterion. × = Does not satisfy the criterion. — = Not reported.

No existing system integrates all of these capabilities into a single deployable solution. Furthermore, current form-agnostic repetition counting systems report only the total number of repetitions, without indicating how many were performed with acceptable biomechanics. As a result, a reported "**10 repetitions**" provides no information about the actual quality, safety, or effectiveness of exercise execution.

4. Problem Statement

Current vision-based exercise monitoring systems primarily focus on **exercise recognition** and **repetition counting**, treating every detected repetition equally regardless of whether it was performed with correct or incorrect biomechanics. Consequently, these systems measure exercise quantity rather than execution quality, providing no indication of whether a repetition was performed safely or effectively.

Although automated exercise assessment has received increasing attention, existing approaches suffer from significant practical limitations. Many rely on **multi-camera configurations**, **depth sensors**, or **wearable devices** such as inertial measurement units (IMUs) or surface electromyography (sEMG), increasing deployment cost and reducing usability in commercial gym environments. Other single-camera approaches typically provide only **binary correct/incorrect classifications** or a single overall

quality score, without identifying the specific biomechanical errors responsible for poor execution or generating personalized corrective feedback.

Moreover, most existing systems have been evaluated under **controlled laboratory settings** or on constrained datasets, limiting their robustness in realistic gym environments characterized by varying camera viewpoints, lighting conditions, occlusions, and multiple people exercising simultaneously. As a result, they cannot reliably support real-time coaching in everyday fitness facilities.

Consequently, users receive little information beyond the total number of completed repetitions. A workout reported as **10 repetitions** provides no indication of **how many repetitions were executed with acceptable technique**, which repetitions contained biomechanical errors, how severe those errors were, or how they should be corrected. This limits both injury prevention and training effectiveness.

Therefore, **there is currently no real-time system that simultaneously**:

- recognizes resistance-training exercises,
- counts repetitions,
- detects specific biomechanical form errors for every repetition,
- distinguishes between valid and invalid repetitions,
- generates explainable corrective feedback,
- operates using only a single ordinary RGB camera without wearable or depth sensors,
- and is validated under realistic, unconstrained gym conditions.

Addressing this gap is essential for developing an intelligent, low-cost, and deployable AI fitness assistant capable of improving exercise safety, movement quality, and overall training effectiveness.

5. Research Objectives

The primary objective of this research is to develop a **real-time, vision-based intelligent fitness assistant** capable of monitoring resistance-training exercises using only a single ordinary RGB camera. Unlike existing systems that focus solely on exercise recognition or repetition counting, the proposed system aims to evaluate the **quality of movement execution** by detecting biomechanical form errors, providing explainable corrective feedback, and distinguishing between correctly and incorrectly performed repetitions.

To achieve this objective, the proposed research seeks to:

- Develop a robust real-time pipeline for recognizing resistance-training exercises from monocular RGB video.
- Accurately count exercise repetitions in real time under realistic gym conditions.
- Detect exercise-specific biomechanical form errors on a per-repetition basis using human pose estimation.
- Generate clear, explainable, and actionable corrective feedback to help users improve exercise technique.
- Differentiate between **valid** and **invalid** repetitions based on predefined biomechanical criteria.
- Validate the proposed framework in unconstrained gym environments without requiring wearable sensors, depth cameras, or multi-camera systems.

6. Research Contributions

The expected contributions of this work are summarized as follows:

C1. A unified single-camera framework for simultaneous exercise recognition, repetition counting, and biomechanical form assessment in real time.

C2. An explainable form-analysis module capable of identifying exercise-specific biomechanical errors rather than providing only binary quality labels or numerical scores.

C3. A validity-aware repetition counting mechanism that distinguishes between correctly and incorrectly executed repetitions instead of reporting only the total repetition count.

C4. A real-time feedback generation module that produces human-readable corrective recommendations immediately after detecting movement errors.

C5. A practical, low-cost solution that requires only a standard RGB camera, eliminating the need for wearable sensors, depth cameras, or multi-camera systems.

C6. Validation of the proposed framework using realistic gym recordings captured under unconstrained conditions, demonstrating its applicability to commercial fitness environments.

C7. Integration of a Large Language Model (LLM) to translate detected biomechanical errors into personalized, natural-language coaching instructions that are understandable by non-expert users.

7. Research Questions and Objectives

The proposed research aims to investigate whether a single-camera computer vision system can provide real-time, explainable, and biomechanically meaningful feedback for resistance-training exercises under realistic gym conditions. The study is guided by the following research questions:

RQ1 — Exercise Recognition and Repetition Counting (Foundation)

Research Question: Can resistance-training exercises and repetition boundaries be detected reliably in real time using a single RGB camera under unconstrained gym environments?

Objective: Develop a real-time vision-based pipeline capable of identifying the performed exercise and accurately counting repetitions as a foundation for higher-level form analysis.

Expected Outcome: A robust exercise recognition and repetition counting module that operates using ordinary camera input without wearable or depth sensors.

RQ2 — Per-Repetition Biomechanical Error Detection (Core Contribution)

Research Question: Can specific biomechanical form errors, such as knee valgus, spinal rounding, insufficient depth, and improper joint alignment, be detected from monocular pose estimation without requiring multi-camera systems or additional sensors?

Objective: Design a form-analysis module capable of extracting biomechanical indicators from human pose landmarks and identifying specific execution errors for each individual repetition.

Expected Outcome: A system that does not only determine whether a repetition is incorrect, but also identifies *what* error occurred and *where* it occurred.

RQ3 — Explainable Corrective Feedback Generation (Core Contribution)

Research Question: Can detected biomechanical errors be converted into understandable and actionable feedback that helps users immediately correct their movement?

Objective: Develop an explainable feedback mechanism that maps detected errors into human-readable recommendations.

Expected Outcome: Instead of outputting "Incorrect repetition," the system provides feedback such as: *"Keep your knees aligned with your toes during the squat to avoid inward collapse."*

RQ4 — Validity-Aware Repetition Counting (Core Contribution)

Research Question: Does distinguishing between valid and invalid repetitions provide a more meaningful training metric than traditional repetition counting?

Objective: Introduce a repetition-quality metric that separates technically correct repetitions from repetitions performed with unsafe biomechanics.

Expected Outcome: Instead of reporting "10 repetitions completed," the system reports:

- 10 total repetitions
- 7 valid repetitions
- 3 repetitions requiring correction

8. Proposed System Overview / Framework

The proposed system introduces a real-time, single-camera AI fitness assistant capable of recognizing exercises, evaluating execution quality, and providing explainable feedback without requiring wearable devices or specialized hardware. The overall pipeline consists of five main stages.

Stage 1 — Video Acquisition

A standard RGB camera captures real-time exercise videos from the gym environment.

Input:

- RGB video stream
- No depth information
- No wearable sensors
- No EMG signals

The system is designed to handle realistic conditions including different camera viewpoints, background variations, lighting changes, and multiple gym users.

Stage 2 — Human Pose Estimation

The captured frames are processed using a monocular human pose estimation model to extract body keypoints. The extracted landmarks include the shoulders, elbows, wrists, hips, knees, and ankles. These landmarks provide the geometric representation required for biomechanical analysis.

Stage 3 — Exercise Recognition and Repetition Detection

Temporal modeling is applied to identify the exercise type, movement phases, and repetition boundaries. For example, a squat movement's pose sequence is passed through a temporal model, yielding a detected repetition.

Stage 4 — Biomechanical Form Analysis

For every detected repetition, the system analyzes biomechanical indicators such as:

Squat: knee valgus, insufficient depth, excessive forward lean.

Deadlift: spinal rounding, incorrect hip position, bar path deviation.

Bench Press: excessive elbow flare, incorrect shoulder position.

The repetition receives an error type, a severity level, and a valid/invalid classification.

Stage 5 — Explainable Feedback Generation

The detected errors are converted into actionable feedback. For example, a detected knee angle deviation is translated into: *"Your knees are moving inward during the squat. Push your knees outward to maintain proper alignment."*

The final system output includes the exercise name, total repetitions, valid repetitions, detected errors, and corrective instructions.

9. Expected Contributions

The main contributions of this research are:

- **A single-camera real-time exercise assessment framework** suitable for practical gym environments.
- **Biomechanically informed error detection** capable of identifying specific execution mistakes rather than only classifying movements as correct or incorrect.
- **Explainable corrective feedback generation** that translates computer vision predictions into understandable coaching instructions.
- **Validity-aware repetition counting**, distinguishing technically correct repetitions from unsafe repetitions.
- **A wearable-free and low-cost solution** that improves accessibility compared with existing multimodal systems.

References

[1] N. Cridlig, *Real-Time Exercise Recognition and Repetition Counting from Monocular Gym Video*, Master's Thesis, Department of Computer Science and Engineering, Alma Mater Studiorum – Università di Bologna, 2026. Available: https://amsdottorato.unibo.it/

[2] R. Khurana, K. Ahuja, Z. Yu, J. Mankoff, C. Harrison, and M. Goel, "GymCam: Detecting, Recognizing and Tracking Simultaneous Exercises in Unconstrained Scenes," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, vol. 2, no. 4, pp. 1–17, 2018. DOI: https://doi.org/10.1145/3287046

[3] P. Parmar, A. Gharat, and H. Rhodin, "Domain Knowledge-Informed Self-Supervised Representations for Workout Form Assessment," *European Conference on Computer Vision (ECCV)*, pp. 105–123, 2022. DOI: https://doi.org/10.1007/978-3-031-19772-7_7

[4] J. Qi, S. Fan, Z. Qiao, et al., "Chain-of-Thought Based Explainable Fitness Assessment (CoT-AFA)," 2025. Available: https://arxiv.org/

[5] Y. Yin et al., "FLEX: Explainable Fitness Assessment via Fitness Knowledge Graph and Multimodal Biomechanical Analysis," 2025. Available: https://arxiv.org/

[6] X. Xu et al., "M3GYM: A Multi-view Multi-modal Dataset for Real-world Gym Exercise Assessment," 2025. Available: https://arxiv.org/

[7] V. Bazarevsky et al., "BlazePose: On-device Real-time Body Pose Tracking," Google Research, 2020. Available: https://arxiv.org/abs/2006.10204

[8] J. Shotton, A. Fitzgibbon, M. Cook, T. Sharp, M. Finocchio, R. Moore, A. Kipman, and A. Blake, "Real-Time Human Pose Recognition in Parts from Single Depth Images," *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1297–1304, 2011. DOI: https://doi.org/10.1109/CVPR.2011.5995316

[9] K. Golshani, M. E. Cinque, P. O'Halloran, K. Softness, L. Keeling, and J. R. Macdonell, "Upper Extremity Weightlifting Injuries: Diagnosis and Management," *Journal of Orthopaedics*, vol. 15, no. 1, pp. 24–27, 2018. DOI: https://doi.org/10.1016/j.jor.2017.11.005

[10] J. W. Keogh and P. W. Winwood, "The Epidemiology of Injuries Across the Weight-Training Sports," *Sports Medicine*, vol. 47, pp. 479–501, 2017. DOI: https://doi.org/10.1007/s40279-016-0575-0

[11] Z. Y. Kerr, C. L. Collins, and R. D. Comstock, "Epidemiology of Weight Training-Related Injuries Presenting to United States Emergency Departments, 1990 to 2007," *American Journal of Sports Medicine*, vol. 38, no. 4, pp. 765–771, 2010. DOI: https://doi.org/10.1177/0363546509351560

[12] M. E. Lavallee and T. Balam, "An Overview of Strength Training Injuries: Acute and Chronic," *Current Sports Medicine Reports*, vol. 9, no. 5, pp. 307–313, 2010. DOI: https://doi.org/10.1249/JSR.0b013e3181f3ed6d

[13] A. Capecci, M. G. Ceravolo, F. Ferracuti, et al., "KIMORE Dataset: Kinect-based Movement Recognition for Rehabilitation Exercises," 2019. Available: https://www.ncbi.nlm.nih.gov/

[14] A. Vakanski, H. Jun, D. Paul, and R. Baker, "A Data Set for Human Motion Analysis in Physical Rehabilitation," *Proceedings of the IEEE International Conference on Healthcare Informatics*, 2018. Available: https://ieeexplore.ieee.org/

[15] S. Chen, J. Qi, S. Fan, Z. Qiao, J. C. Yeo, and C. T. Lim, "Deep Learning in Human Activity Recognition with Wearable Sensors: A Review on Advances," *Sensors*, vol. 22, no. 4, 2022. DOI:

https://doi.org/10.3390/s22041476